Group Theory
With Okron's Templates

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Contents

1. Introduction

Note. 1.1
All counters (Theorem, Definition, Proposition, etc.) are automatically reset at each section heading (level 1).

1.1. Group Initiation

Def. 1.1 *(Group)*
A **group** is an ordered pair $(G, *)$ where G is a set and $* : G \times G \to G$ is a binary operation such that:
1. **Associativity:** $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$.
2. **Identity:** There exists $e \in G$ such that $a * e = e * a = a$ for all $a \in G$.
3. **Inverse:** For each $a \in G$ there exists $a^{\{-1\}} \in G$ with $a * a^{\{-1\}} = a^{\{-1\}} * a = e$.

With the previous definition we are now able to make some observations, as the following:

Prop. 1.1 *(Uniqueness of identity and inverses)*
In a group $(G, *)$ the identity element is unique, and every element has a unique inverse.

And also we can talk about Lemmas.

Lemma. 1.1 *(Powers in a group)*
For any $a \in G$ and $m, n \in \mathbb{Z}$, we have $a^m a^n = a^{m+n}$ and $(a^m)^n = a^{mn}$, where a^k is defined recursively.

Finally it may be useful to write some Affirmations.

Aff. 1.1 *(Left and right cancellation)*
In a group, $ab = ac$ implies $b = c$, and $ba = ca$ implies $b = c$.

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2. Theorems and Proofs

∯ **Theorem. 2.1** *(Lagrange's Theorem)*
If G is a finite group and $H \leq G$ a subgroup, then $|H|$ divides $|G|$. In particular, $|G| = [G : H], |H|$.

∯ **Corollary. 2.1** *(Order of an element)*
For any element $g \in G$ of finite order, $|g|$ divides $|G|$.

∯ ***Proof.* 2.1**
Let $n = |g|$ and let $H = \langle g \rangle$ be the cyclic subgroup generated by g. Then $|H| = n$. By Lagrange's theorem n divides $|G|$.

■

∯ **Remark. 2.1** The converse of Lagrange's theorem does not hold in general. For example, A_4 has order 12 but has no subgroup of order 6.

3. Examples and Exercises

∯ **Example. 3.1** *(Cyclic group)*
The group $\mathbb{Z}_n$ under addition modulo n is cyclic. Every subgroup of a cyclic group is also cyclic.

∯ **Exercise. 3.1** *(A non-abelian group)*
Show that the symmetric group S_3 is not abelian.

∯ ***Proof.* 3.1**
S_3 consists of permutations of three elements. Compute $(12)(13) = (132)$ but $(13)(12) = (123)$. Since $(132) \neq (123)$, the group is non-abelian.

■

∯ **Exercise. 3.2** *(Subgroup test)*
Prove that a non-empty subset $H \subseteq G$ is a subgroup iff $ab^{-1} \in H$ whenever $a, b \in H$.

4. Figures, Code and Plots

Figure 1: The best profile picture ever.

If the image is missing, you can still see the figure structure.

$\oint$ **Code. 4.1** *(Python Multiplication)*

```python
def multiply(a, b):
    return a * b

result = multiply(3, 5)
print(f"3 * 5 = {result}")
```

$\longrightarrow$ **`stdout:`**

```
3 * 5 = 15
```

This is a practical way to include code snipets on your writings.